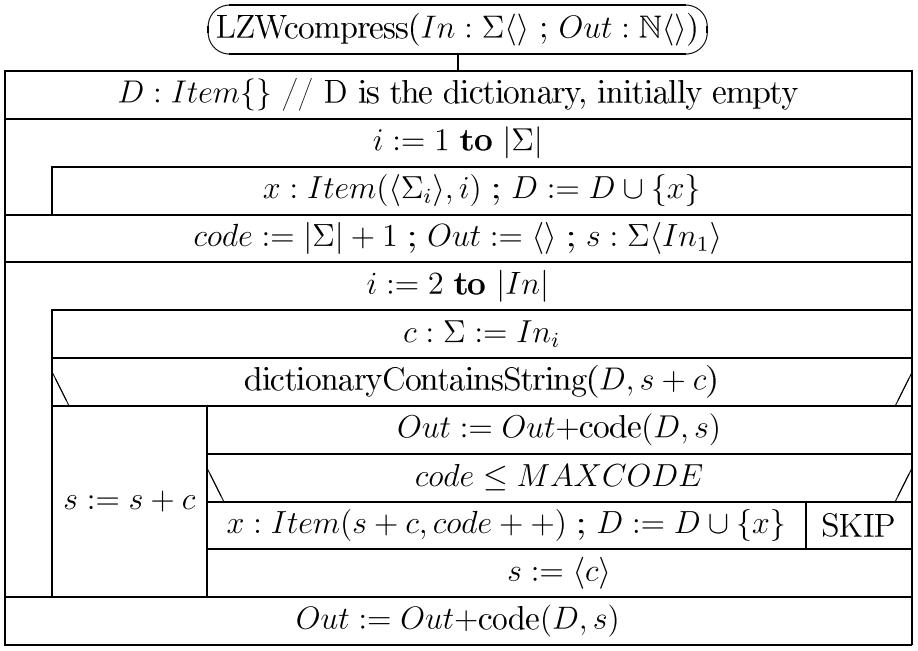
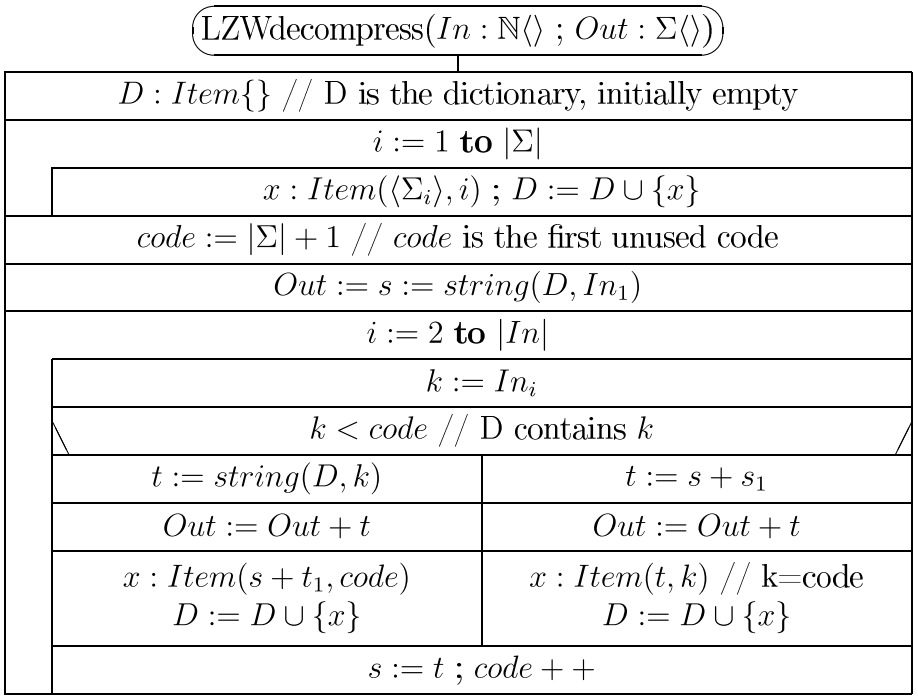

LZWcompress($In : \Sigma\langle\rangle$; $Out : \mathbb{N}\langle\rangle$)
$D : Item\{\}$ // D is the dictionary, initially empty
$i := 1$ **to** $|\Sigma|$
$x : Item(\langle\Sigma_i\rangle, i)$; $D := D \cup \{x\}$
$code := |\Sigma| + 1$; $Out := \langle\rangle$; $s : \Sigma\langle In_1\rangle$
$i := 2$ **to** $|In|$
$c : \Sigma := In_i$
dictionaryContainsString($D, s + c$)
$s := s + c$
$Out := Out$+code(D, s)
$code \leq MAXCODE$
$x : Item(s + c, code + +)$; $D := D \cup \{x\}$
SKIP
$s := \langle c\rangle$
$Out := Out$+code(D, s)


LZWdecompress($In : \mathbb{N}\langle\rangle$; $Out : \Sigma\langle\rangle$)
$D : Item\{\}$ // D is the dictionary, initially empty
$i := 1$ **to** $|\Sigma|$
$x : Item(\langle\Sigma_i\rangle, i)$; $D := D \cup \{x\}$
$code := |\Sigma| + 1$ // $code$ is the first unused code
$Out := s := string(D, In_1)$
$i := 2$ **to** $|In|$
$k := In_i$
$k < code$ // D contains k
$t := string(D, k)$
$t := s + s_1$
$Out := Out + t$
$Out := Out + t$
$x : Item(s + t_1, code)$
$D := D \cup \{x\}$
$x : Item(t, k)$ // k=code
$D := D \cup \{x\}$
$s := t$; $code + +$